

Stem-and-leaf plots of statistical analysis also rely on micro/macro design. Each data point simultaneously states its value and fills a space representing one counted unit, like the names on the Vietnam Veterans Memorial, with those spaces in turn assembling to form a profile of the overall univariate distribution. Envisioned here are the heights of 218 volcanoes; each individual number helps to build the histogram. Micro-

0|9 = 900 feet

Stem-and-leaf displays:
heights of 218 volcanoes, unit 100 feet.

19|3 = 19,300 feet

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0 98766562
1 97719630
2 6998776654422211009850
3 876655412099551426
4 9998844331929433361107
5 97666666554422210097731
6 898665441077761065
7 98855431100652108073
8 633322122937
9 377655421000493
10 0984433165212
11 4965201631
12 4521164
13 47830
14 00
15 676
16 52
17 92
18 5
19 39730

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data has replaced the information-empty bars of a traditional barchart. This idea of making each graphical element *repeatedly* effective animated design of the stem-and-leaf plot. In describing his invention, John Tukey wrote: "If we are going to make a mark, it may as well be a meaningful one. The simplest—and most useful—meaningful mark is a digit."⁷

In a similar fashion, this train schedule below positions the individual departure times so that they add up to a frequency distribution. For trains that run often, leading hour-digits need not be repeated over and over, and, instead, minutes can be stacked:

平日下り

時	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
分	06	18	31	46	58	11	21	30	41	49	59	10	20	30	40	50	58	58	58	58
乗車駅	06	18	31	46	58	11	21	30	41	49	59	10	20	30	40	50	58	58	58	58
乗車駅	08	12	00	09	20	22	32	42	52	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
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乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
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乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58
乗車駅	08	12	09	17	28	32	42	52	58	58	58	58	58	58	58	58	58	58	58	58</

⁷ John W. Tukey, "Some Graphic and Semigraphic Displays," in T. A. Bancroft, ed., *Statistical Papers in Honor of George W. Snedecor* (Ames, Iowa, 1972), 296.

Keihin Express Line at Yokohama Station, Sagami Tetsudo Company, 1985 timetable, 76. Encodings indicate types of trains (super express, commuter, and so on) and various local stops.

Reported is the overall time distribution of 292 daily trains, with peaks during morning and evening rush hours. The shrewd design saves 777 characters, avoiding this typographical extravaganza below, which lacks the intensive annotation of the stem-and-leaf original and also fails to provide clear testimony about frequency of train service by hour.⁸

⁸ The stem-and-leaf schedule contains 619 numbers; the typographic version 1,396 numbers and periods. Thus the stem-and-leaf schedule saves 777 characters, and, more importantly, gives a much a better sense of comparison of train times.

5.06	7.17	8.28	9.31	10.40	11.57	13.12	14.28	15.45	16.52	17.53	18.45	19.40	20.39	21.51	23.36
5.18	7.23	8.30	9.33	10.45	11.59	13.17	14.32	15.48	16.59	17.55	18.48	19.43	20.41	21.56	23.47
5.31	7.26	8.32	9.41	10.49	12.05	13.19	14.37	15.52	17.01	17.57	18.53	19.45	20.46	21.01	23.54
5.40	7.30	8.38	9.43	10.54	12.08	13.25	14.39	15.57	17.04	18.01	18.55	19.47	20.50	22.09	24.03
5.46	7.36	8.40	9.50	10.57	12.12	13.28	14.45	15.59	17.10	18.03	18.57	19.51	20.52	22.11	24.15
5.58	7.38	8.42	9.53	11.00	12.17	13.32	14.48	16.05	17.12	18.05	18.01	19.53	20.58	22.17	24.21
6.04	7.40	8.50	9.57	11.05	12.19	13.37	14.52	16.08	17.14	18.07	19.04	19.55	21.01	22.21	24.23
6.12	7.45	8.52	10.01	11.08	12.25	13.39	14.57	16.09	17.19	18.13	19.06	20.00	21.06	22.29	
6.18	7.47	8.54	10.03	11.12	12.28	13.45	14.59	16.16	17.22	18.15	19.08	20.02	21.09	22.32	
6.21	7.49	9.00	10.07	11.17	12.32	13.48	15.05	16.18	17.24	18.17	19.13	20.04	21.11	22.39	
6.30	7.54	9.02	10.11	11.19	12.37	13.52	15.08	16.21	17.26	18.21	19.15	20.10	21.18	22.44	
6.38	7.56	9.04	10.12	11.25	12.39	13.57	15.12	16.27	17.30	18.23	19.17	20.12	21.21	22.51	
6.41	7.58	9.10	10.17	11.28	12.45	13.59	15.17	16.29	17.32	18.26	19.20	20.14	21.26	22.53	
6.49	8.03	9.12	10.20	11.32	12.48	14.05	15.19	16.32	17.34	18.28	19.23	20.19	21.29	22.59	
6.55	8.06	9.14	10.22	11.37	12.52	14.08	15.25	16.38	17.36	18.33	19.25	20.21	21.31	23.04	
6.59	8.09	9.20	10.26	11.39	12.57	14.12	15.28	16.40	17.40	18.35	19.27	20.23	21.36	23.10	
7.03	8.18	9.22	10.29	11.45	12.59	14.17	15.32	16.42	17.43	18.37	19.32	20.30	21.41	23.14	
7.08	8.20	9.24	10.34	11.48	13.05	14.19	15.37	16.48	17.45	18.41	19.34	20.32	21.46	23.21	
7.14	8.22	9.29	10.37	11.52	13.08	14.25	15.39	16.50	17.47	18.43	19.36	20.34	21.50	23.30	

In all these micro/macro designs, the same ink serves more than one informational purpose; graphical elements are multifunctioning. This suggests a missed opportunity in the stem-and-leaf timetable—surely leaves of numbers can grow from *both* sides of a central stem. And so it is; the finely detailed timetable below records trains running in several directions from the station, with the platforms 7-8 at left and platforms 5-6 at right (at the arrows, note how numbers serpentine around a bend when times for the morning rush hour exceed the grid). Sometimes this arrangement is called a "back to back stem-and-leaf plot." Nonetheless, Japanese train passengers have managed to use the schedules for decades without ever knowing the fancy name.

7・8 5・6

横浜駅発東海道線

59	09	4	54	25	4	52
58	09	4	54	25	4	52
57	09	4	54	25	4	52
56	09	4	54	25	4	52
55	09	4	54	25	4	52
54	09	4	54	25	4	52
53	09	4	54	25	4	52
52	09	4	54	25	4	52
51	09	4	54	25	4	52
50	09	4	54	25	4	52
49	09	4	54	25	4	52
48	09	4	54	25	4	52
47	09	4	54	25	4	52
46	09	4	54	25	4	52
45	09	4	54	25	4	52
44	09	4	54	25	4	52
43	09	4	54	25	4	52
42	09	4	54	25	4	52
41	09	4	54	25	4	52
40	09	4	54	25	4	52
39	09	4	54	25	4	52
38	09	4	54	25	4	52
37	09	4	54	25	4	52
36	09	4	54	25	4	52
35	09	4	54	25	4	52
34	09	4	54	25	4	52
33	09	4	54	25	4	52
32	09	4	54	25	4	52
31	09	4	54	25	4	52
30	09	4	54	25	4	52
29	09	4	54	25	4	52
28	09	4	54	25	4	52
27	09	4	54	25	4	52
26	09	4	54	25	4	52
25	09	4	54	25	4	52
24	09	4	54	25	4	52
23	09	4	54	25	4	52
22	09	4	54	25	4	52
21	09	4	54	25	4	52
20	09	4	54	25	4	52
19	09	4	54	25	4	52
18	09	4	54	25	4	52
17	09	4	54	25	4	52
16	09	4	54	25	4	52
15	09	4	54	25	4	52
14	09	4	54	25	4	52
13	09	4	54	25	4	52
12	09	4	54	25	4	52
11	09	4	54	25	4	52
10	09	4	54	25	4	52
9	09	4	54	25	4	52
8	09	4	54	25	4	52
7	09	4	54	25	4	52
6	09	4	54	25	4	52
5	09	4	54	25	4	52

Tokaido Line at Yokohama Station, Sagami Tetsudo Company, 1985 timetable, 72.